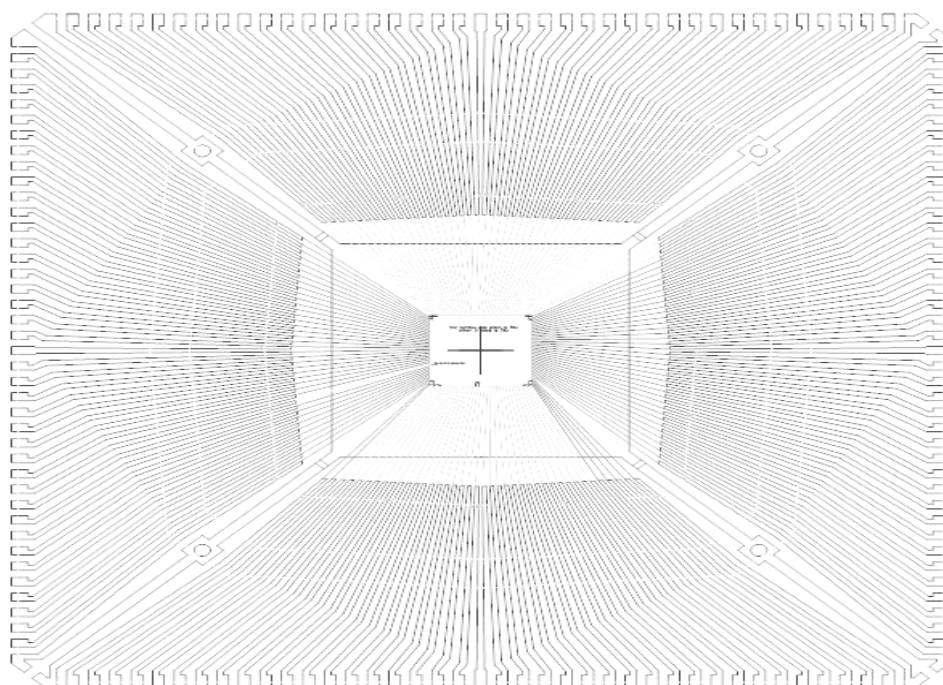


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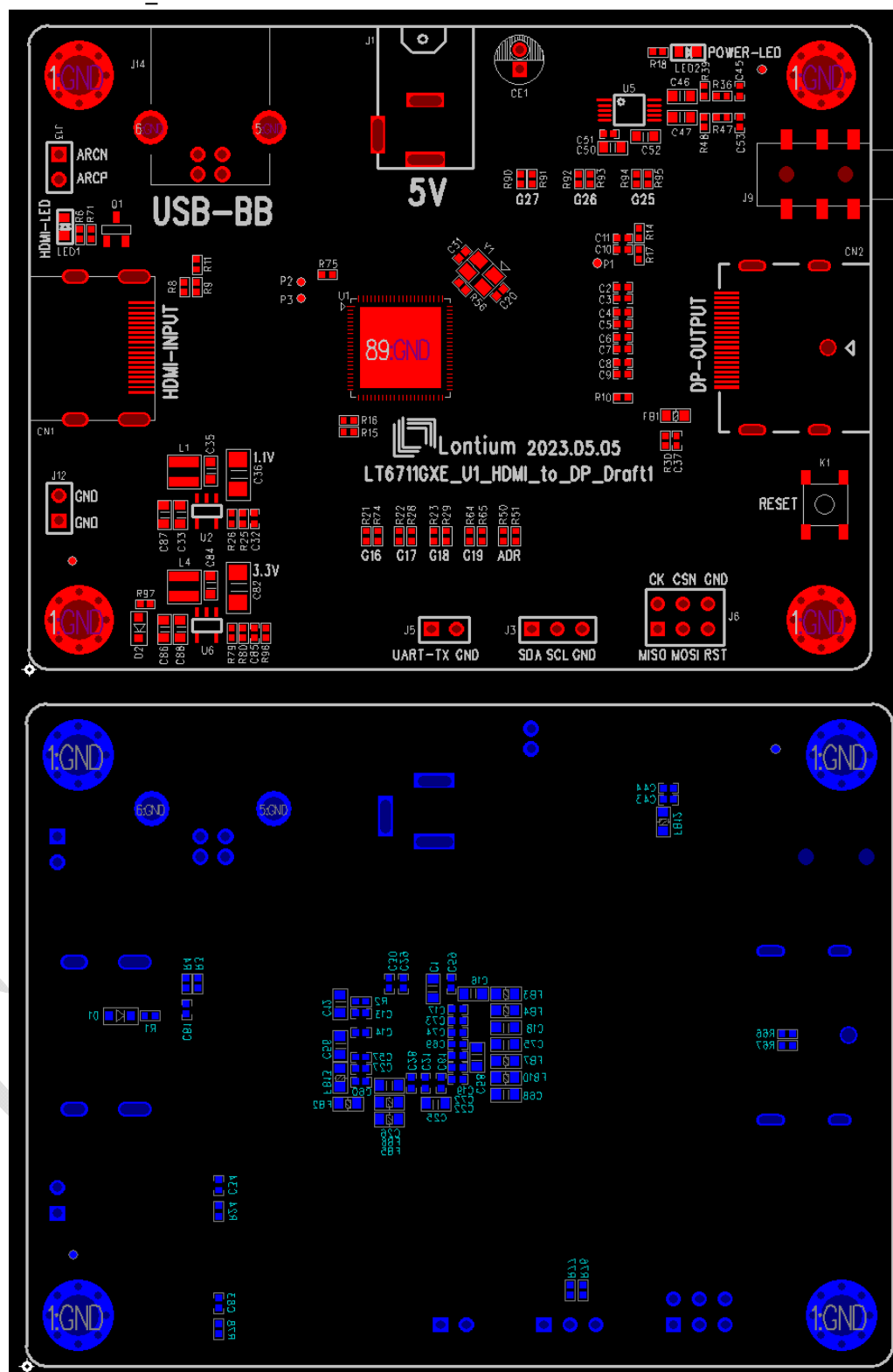
LT6711GXE_HDMI to DP
Board_User_Guide

龙迅半导体

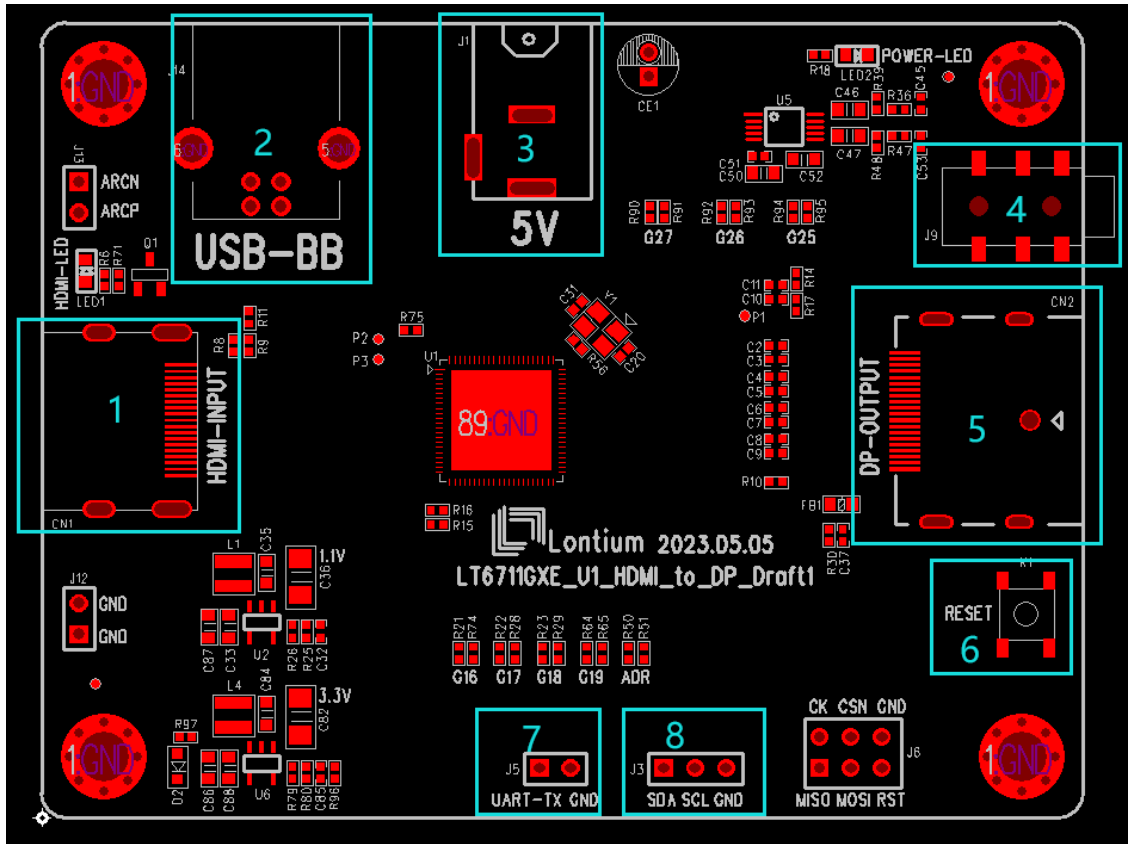


We produce mixed-signal products for a better digital world!

1 LT6711GXE HDMI to DP Demo Board



2 说明(Description)



3

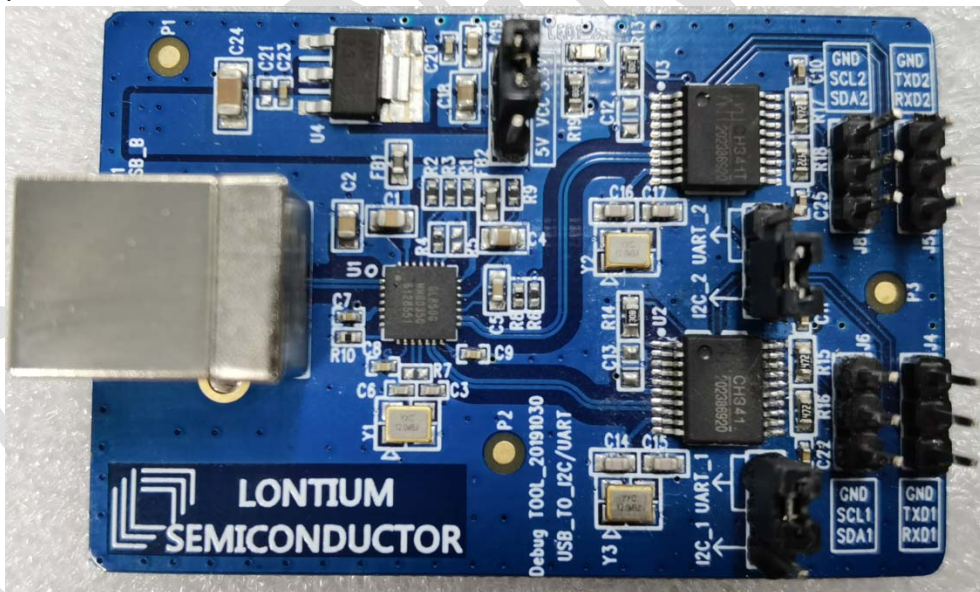
如上图所示(As shown in the figure above):

- ① 是 HDMI 输入端插座，通过一根 HDMI 线连接 HDMI source。(It is HDMI input connector, which should be connected to an HDMI source through an HDMI cable.)
- ② 是 USB-B 插座，转为 BB 使用，用一根 USB 线连接电脑，可以通过 BB 升级芯片程序。(It is a USB-B socket and used for BB. is used to connect the computer. The chip firmware can be upgraded through BB by a USB cable connected to PC)
- ③ 是电源插座，务必接入 5V 电源。(It is power socket, and it must be connected to 5V power supply.)
- ④ 是耳机插口，为耳机声音输出。(It is headphone jack for audio output.)
- ⑤ 是 DP 输出插座，通过一根 DP 线连接 DP sink。(It is DP output connector, which should be connected to a DP sink through a DP cable.)
- ⑥ 是 RESET 按键，按下芯片复位。(It is reset button. Press to reset the chip.)

- ⑦ 是串口打印输出插针，具体串口使用方式见第 3 节。(It is serial port pin header. Please refer to section 3 for usage.)
- ⑧ 是升级程序和 debug 用的 I2C 插针，具体的 I2C 使用方法见第 4 节。(It is I2C pin header for debug and FW update. Please refer to section 4 for usage.)
- ⑨ 其他插针在板上有丝印标注功能。(See marks on the board for other pin headers.)

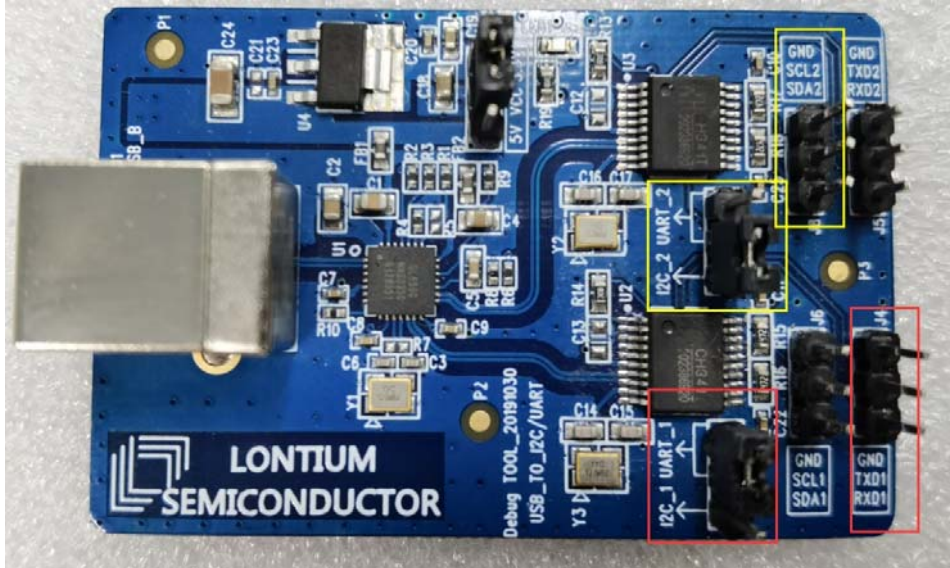
4 Serial Port

- 4.1 串口打印可以显示芯片的一些信息和状态。(Serial port can print some information and status of the chip.)
- 4.2 Debug 工具板如下图所示，这是龙迅制作的 USB2.0 转 I2C+串口的工具板。其中左边接口为 USB-B 型接口，通过 USB 线连接 PC；右边为 I2C 及 UART 串口的插针。(The debug tool board is shown in the figure below. This is a tool board, made by Lontium, to implement USB2.0 to I2C and serial port conversion. The left interface is a USB-B receptacle which should be connected to a PC through a USB cable. The right interface is I2C and UART pin headers.)



注意跳线帽的对应关系，如下图所示，黄色框中的跳线帽连接 I2C_2 部分，所以右上的 GND/SCL2/SDA2 可以被使用，而 GND/TXD2/RXD2 被禁用；同理，红色框中的跳线帽连接 UART_1 部分，所以右下的 GND/TXD1/RXD1 可以被使用，而 GND/SCL1/SDA1 被禁用。(Please note the jumper caps shown in the figure below. The jumper cap in the yellow box is connected to I2C_2, which means GND / SCL2 / SDA2 on the top right should be used while GND / TXD2 / RXD2 is disabled. Similarly, the jumper cap in the red box is

connected to UART_1, which means GND / TXD1 / RXD1 in the bottom right should be used while GND / SCL1 / SDA1 is disabled.)

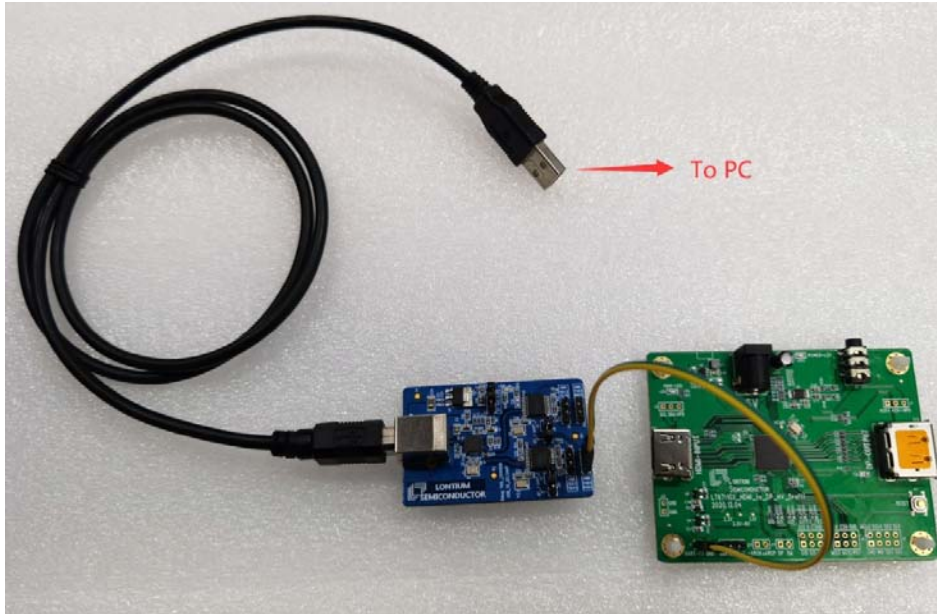


4.3 驱动 (Driver)

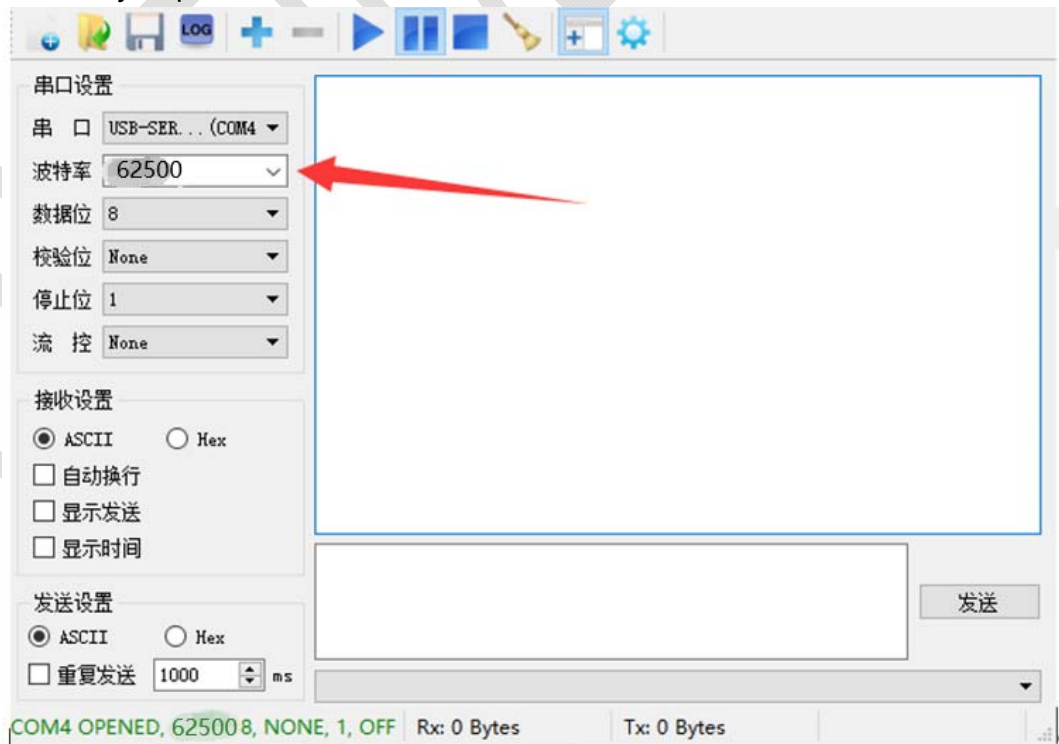
初次把此工具板接入 PC, PC 可能需要安装驱动, 具体驱动程序可以通过网络搜索 CH341。(For the first time, if you connect this tool board to a PC, the PC may need to install drivers. You can search CH341 driver through Ethernet.)

4.4 连接如下图所示 (The connection is shown in the figure below.)

连接 Debug tool board 的 GND/RXD1 与 LT6711GXE 的 GND/UART-TX, 注意, 是 LT6711GXE 的 TX 连接到 Debug tool board 的 RX。(Connect GND / RXD1 of debug tool board to GND / UART-TX of LT6711GXE. Note that TX of LT6711GXE is connected to RX of debug tool board.)



- 4.5 软件：通过网络获取串口打印的应用软件。(Software: You can get the application software of serial port print through Ethernet.)
如 3.3 章节连接好之后，可以通过应用软件在 PC 上看到此端口，设置波特率为 62500，开始运行串口打印。(After connected following section 3.3, you can see this port on PC through application software, set baud rate to 62500, and ready to print.)



4.6 串口打印 (serial port print)

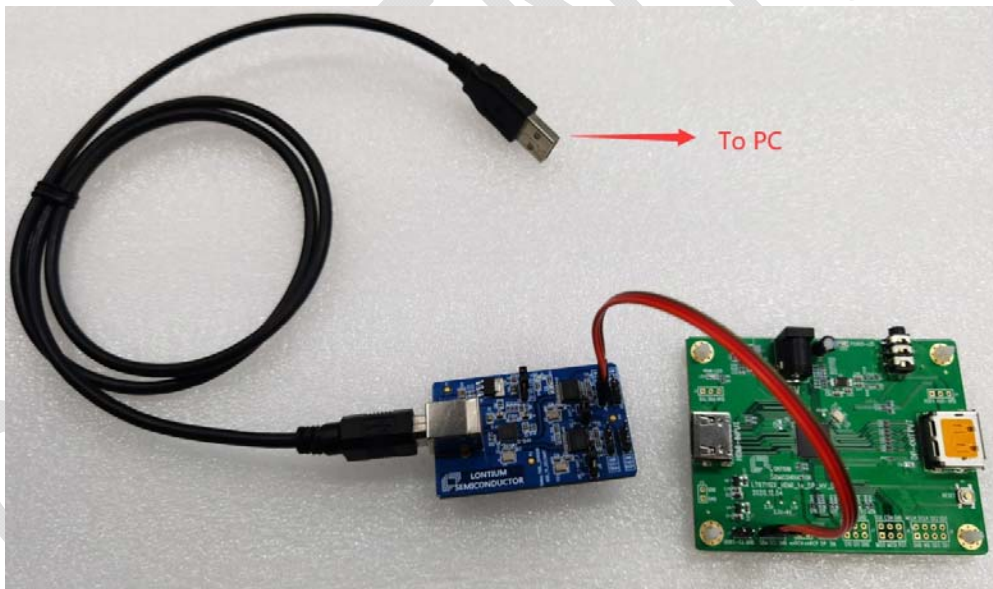
在连接好之后，给 LT6711GXE 板子上电，此时就可以抓取到 LT6711GXE 的串口打印，如下图所示。(Power on the LT6711GXE board, then you can see the serial port printing from LT6711GXE as shown in the figure below.)

5 I2C

5.1 I2C 主要作用是读写寄存器以及给芯片升级程序。(The main function of I2C is to read/write registers and update the firmware of chip.)

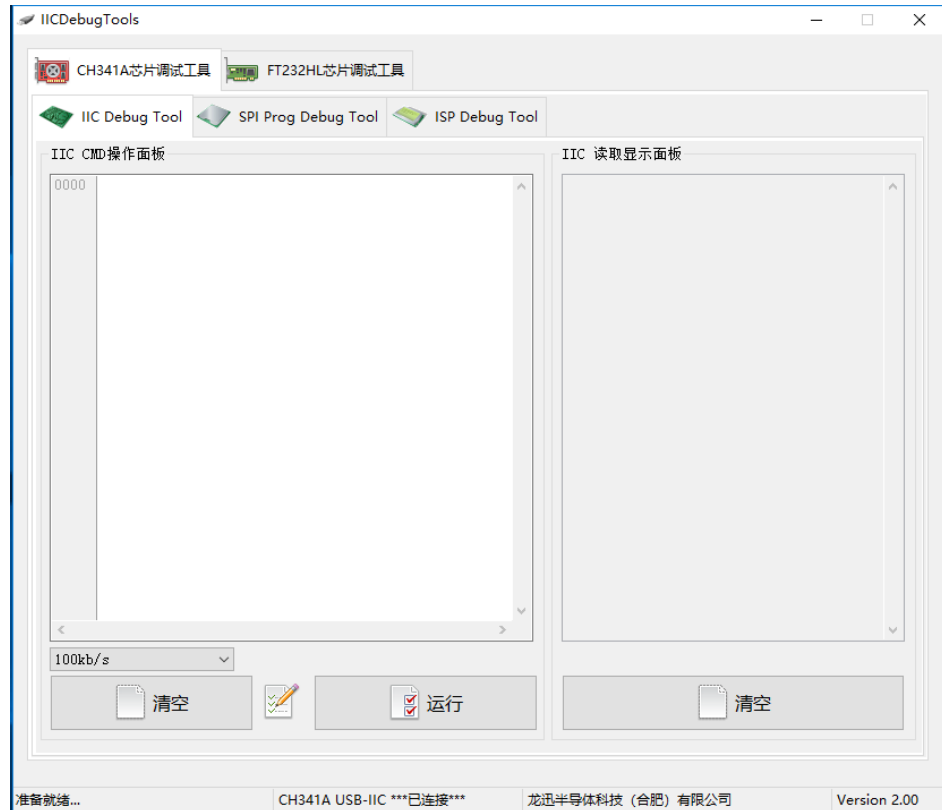
5.2 使用 3.1 和 3.2 章节中的介绍安装驱动。(Use the method in section 3.1 and 3.2 to install the drivers.)

5.3 连接如下图所示 (The connection is shown in the figure below)
连接 Debug tool board 的 GND/SCL2/SDA2 与 LT6711GXE 的 GND/SCL/SDA。(Connect GND / SCL 2 / SDA2 of debug tool board to GND / SCL / SDA of LT6711GXE board.)

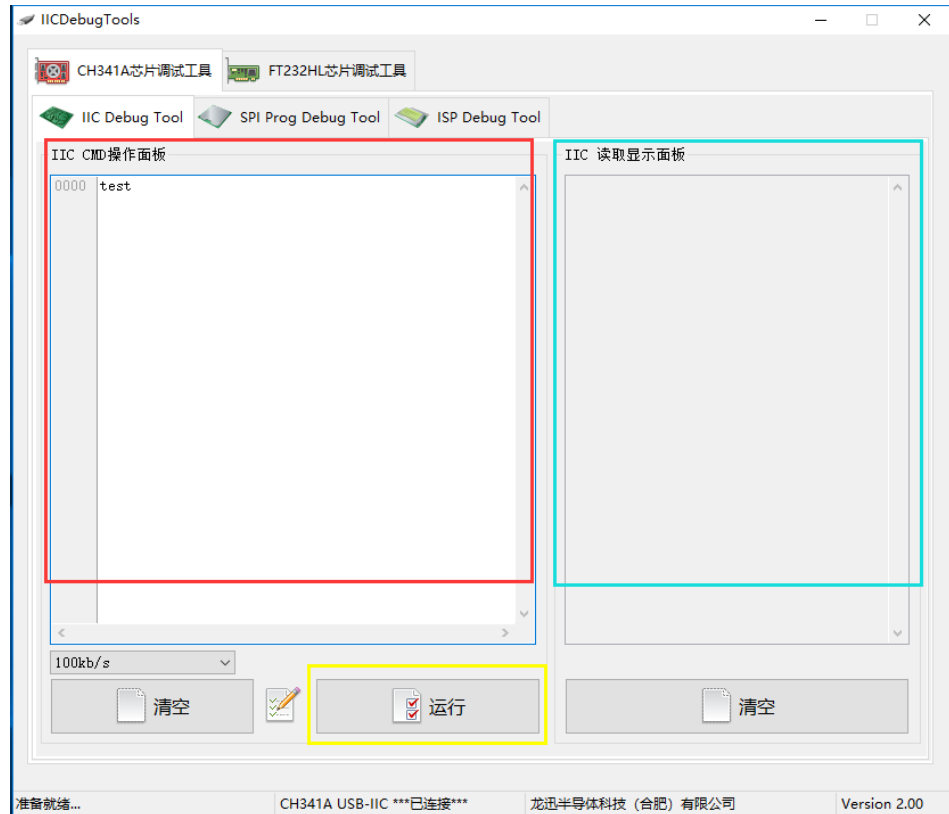


5.4 通过 I2C 读写寄存器 (Read/write registers through I2C)

5.4.1 应用软件需要从龙迅获取，界面如下图所示。(You can get the application software from Lontium, and the interface is shown in the figure below.)

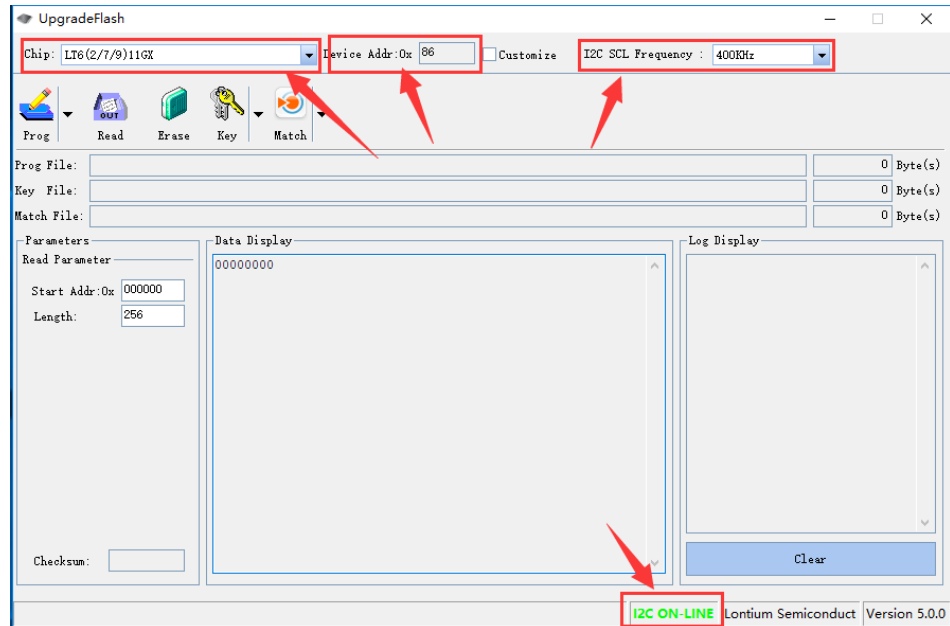


- 5.4.2 读写的寄存器的指令需要从龙迅获取。(You can get the read/write commands from Lontium.)
- 5.4.3 如下图所示，在红色框中输入指令后，点击黄色框中的运行，在蓝色框中会显示读写的结果。(As shown in the figure below, enter commands in the red box, click Run in the yellow box, and the read/write results will be displayed in the blue box.)

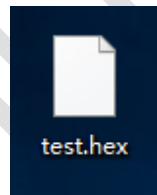


5.5 通过 I2C 烧录芯片程序(flash firmware through I2C)

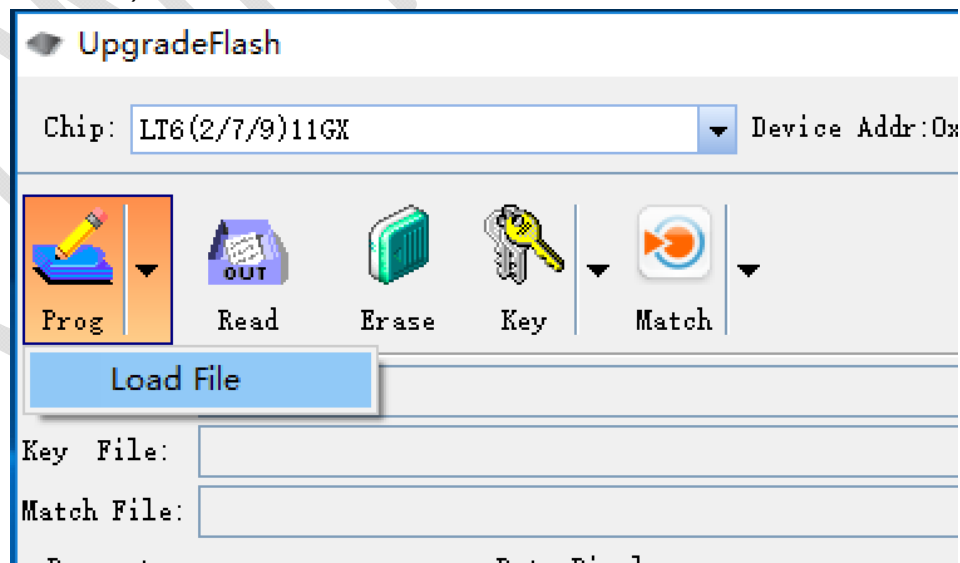
- 5.5.1 应用软件需要从龙迅获取。(You can get the application software from Lontium.)
- 5.5.2 在连接好板子与工具之后，给 LT6711GXE 上电，此时界面如下图所示，注意箭头所指的几个地方是否正确。(After connecting the tool board and LT6711GXE board, power on the LT6711GXE board, the interface is shown in the figure below. Make sure the contents in the red boxes are all correct.)



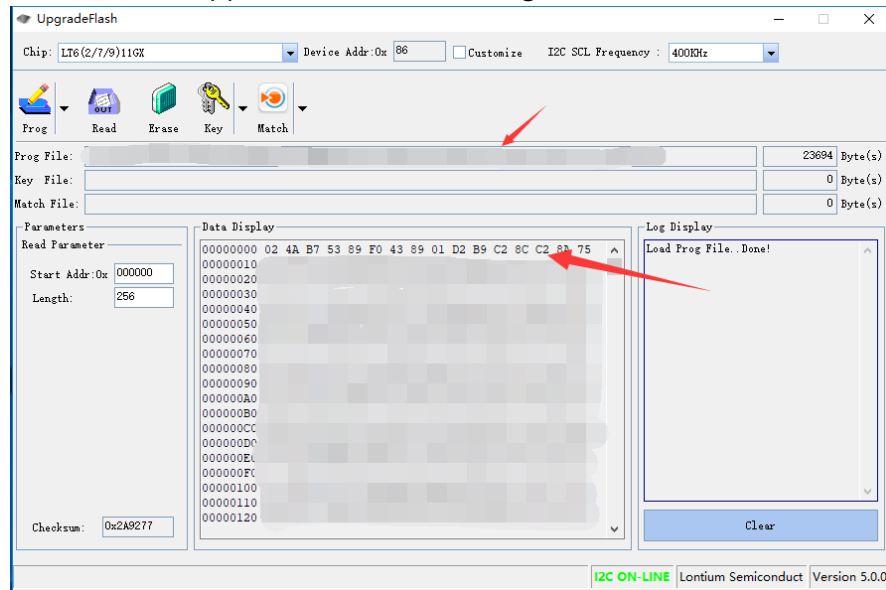
5.5.3 芯片的固件需要从龙迅获取，烧录固件是一个.hex 文件。(You can get the firmware of hex file from Lontium.)



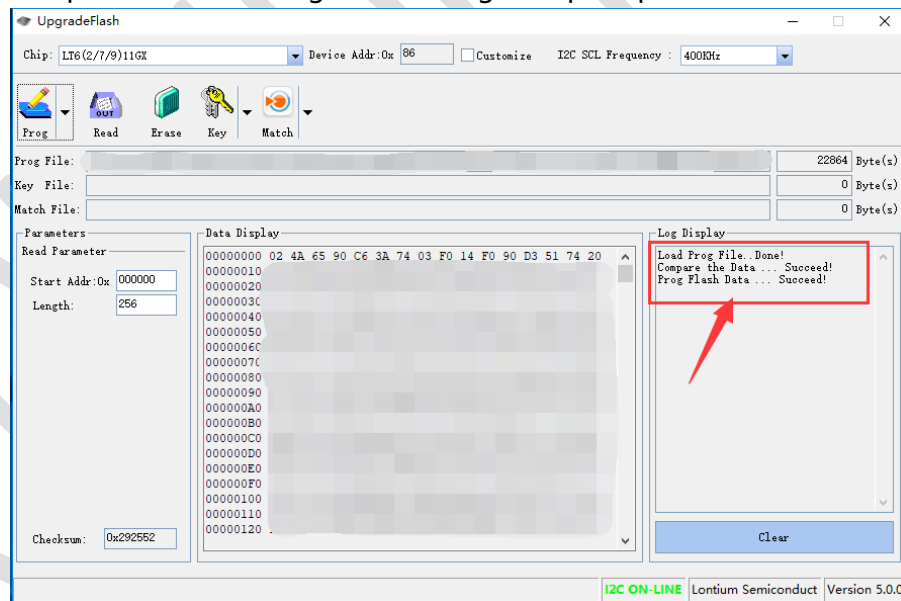
5.5.4 点击“Prog”旁的箭头，点击“Load File”，加载芯片程序的.hex 文件。(Click the arrow next to "Prog" and click "Load File" to load the hex firmware.)



5.5.5 会出现文件的路径以及二进制码，如下图所示。(The path and binary code of the file will appear as shown in the figure below.)



5.5.6 点击“Prog”开始烧录程序，等待进度条结束，如果烧录成功会有如下箭头所指的提示。(Click "Prog" to flash and wait for the progress bar to complete. If the flashing succeed, logs will prompt as indicated below.)



5.5.7 此时已经完成芯片的程序更新，断电重启后新程序开始工作。(After flashing, you just need to power cycle the LT6711GXE board to get the new firmware to work.)

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